

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

November 2013

EE216 Integrated Circuits & Analysis

Date: 16.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Draw the pin diagram of 741 IC. And write down the characteristics of an ideal op-amp. [07]

Q - 2 (a) Draw the different circuit arrangements for obtaining positive and negative supply voltages for an op-amp. [06]

(b) Note down the different feedback configurations for the feedback amplifier. And draw the block diagram for each configuration with signal flow directions. [08]

OR

Q - 2 (a) Find out closed-loop voltage gain (A_F) for voltage-series negative feedback amplifier. [05]

(b) Explain Voltage to current converter using 741 IC with floating load with neat circuit diagram and waveforms. [05]

(c) Draw pin diagram for 555 timer IC. [04]

Q - 3 (a) Explain 555 timer as an astable multivibrator with neat circuit diagram and waveforms. [08]

(b) Using the 7805C voltage regulator, design a current source that will deliver a 0.25 A current to the $48\ \Omega$, 10 W load. [06]

OR

Q - 3 (a) Explain Summing Amplifier with inverting configuration. [05]

(b) Explain first order high-pass butterworth filter with neat circuit diagram and frequency response curve. [07]

(c) Write down output voltage equation for inverting and non-inverting amplifier if its input voltage is V_i . [02]

SECTION – II

Q - 4 Draw and explain block diagram of typical op-amp. Give the classification of ICs. [07]

Q - 5 (a) Define the following terms with necessary diagram. [06]

(i) Input bias current Common-Mode Rejection Ratio (CMRR)

(ii) Input offset Voltage

(iii) Common-Mode Rejection Ratio (CMRR)

(iv) Supply Voltage Rejection Ratio (SVRR)

(b) Write down the features of 741 op-amp IC. [06]

(c) Draw the pin diagram for 78XX and 79XX series voltage regulator ICs. [02]

OR

Q - 5 (a) Explain All-pass filter with neat circuit diagram and output waveforms. [07]

(b) Explain Square-wave generator using 741 op-amp IC with neat circuit diagram and output waveforms. [07]

Q - 6 (a) Explain Integrator with neat diagram and waveforms. [07]

(b) Derive the slew rate equation for 741C. [07]

OR

Q - 6 (a) Explain Zero-crossing detector using 741 IC with its input and output waveforms. [05]

(b) Explain first order low-pass butterworth filter with neat circuit diagram and frequency response curve. [07]

(c) Draw circuit diagram for a Subtractor and write down its output voltage equation. [02]
